॥ श्रीगणेशाय नमः ॥

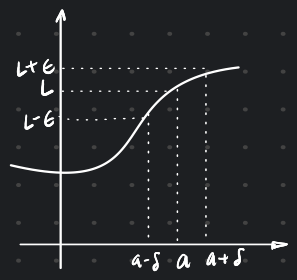
Limits

The limit of a function at a point is defined as a value that the function approaches for the given input values.

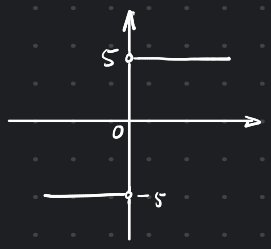
When x tends to move closer to a point a, f(x) will tend to move closer to the value h.

$$\lim_{x \to a} f(x) = h$$

Given $\epsilon > 0$, there exists $\delta > 0$ s.t. if x is within δ of a, then $f(x)$ will lee within ϵ of L.



e.g. $f(x) = \begin{cases} 5, & x > 0 \\ -5, & x < 0 \\ 0, & x = 0 \end{cases}$



→ $\lim_{x \to 0^-} f(x) = -5$

$\lim_{x \to 0^+} f(x) = +5$

$\lim_{x \to 0} f(x)$ does not exist.

Limits can be calculated while x approaching some a from the left of a, as well as the right of a.
If left hand limit at a = right hand limit at a, then limit at a exists.

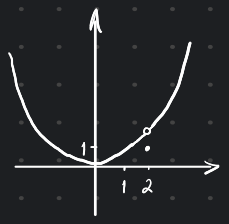
i.e. $\lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) \Rightarrow \lim_{x \to a} f(x)$ exists.

Q. Does $\lim_{x \to 1} f(x)$ exist?

→ $\lim_{x \to 1^-} f(x) = 5$, $\lim_{x \to 1^+} f(x) = 5 \Rightarrow \lim_{x \to 1} f(x) = 5$

$f(1) = 5$

Q. $f(x) = \begin{cases} x^2, & x \neq 2 \\ 1, & x = 2 \end{cases}$



$\lim_{x \to 2} f(x) = ?$ $f(2) = ?$

→ $\lim_{x \to 2^-} f(x) = 4$

$\lim_{x \to 2^+} f(x) = 4$

$\Rightarrow \lim_{x \to 2} f(x) = 4$,

$f(2) = 1$

Q. $f(x) = [x]$, where $[x]$ is the largest integer that is less than or equal to x.

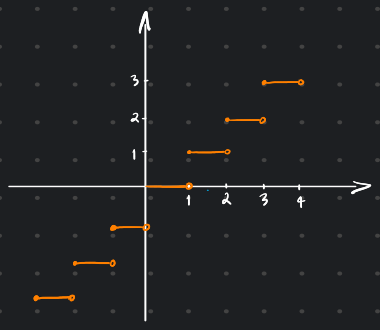
$f(3) = ?$ $\lim_{x \to 3} f(3) = ?$

→ $f(3) = 3$

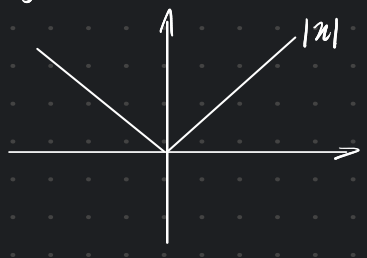
$\lim_{x \to 3^-} = 2$, $\lim_{x \to 3^+} = 3$

$\neq$

$\lim_{x \to 3} f(x)$ does not exist!



Q. $f(x) = |x|$

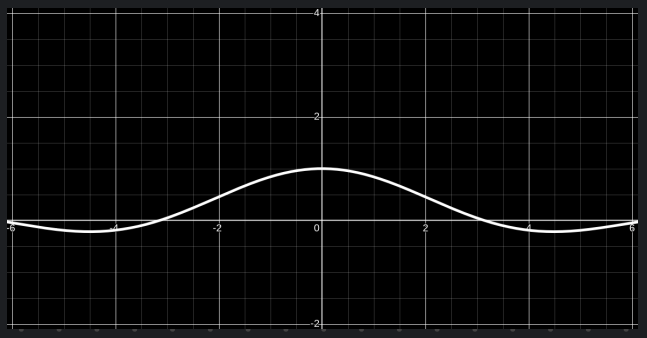


$\lim_{x \to 0^-} f(x) = 0$

$\lim_{x \to 0^+} f(x) = 0$

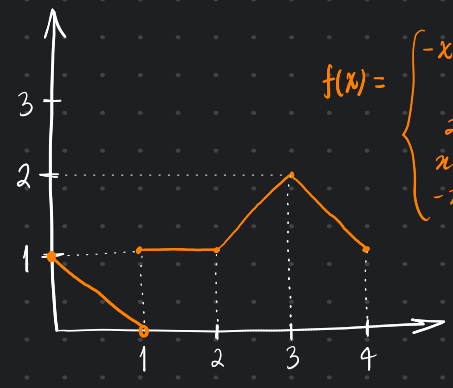
so $\lim_{x \to 0} f(x) = 0$. Also, $f(x) = 0$

8. $f(x) = \frac{\sin x}{x}$



$$\lim_{x \to 0} \frac{\sin x}{x} = 1$$

8.



$$f(x) = \begin{cases} -x+1; & 0 \le x < 1 \\ 1; & 1 \le x < 2 \\ 2; & x = 2 \\ x-1; & 2 < x \le 3 \\ -x+5; & 3 < x \le 4 \end{cases}$$

a) $\lim_{x \to 0^+} f(x) = 1$

b) $\lim_{x \to 1} f(x)$ $\begin{cases} \lim_{x \to 1^-} f(x) = 0 \\ \lim_{x \to 1^+} f(x) = 1 \end{cases}$ $\rightarrow \lim_{x \to 1} f(x)$ does not exist

c) $\lim_{x \to 2} f(x)$ $\begin{cases} \lim_{x \to 2^-} f(x) = 1 \ \checkmark \\ \lim_{x \to 2^+} f(x) = 1 \ \checkmark \end{cases}$

Properties of limits

1. $\lim_{x \to a} k = k$; k is a constant.

2. $\lim_{x \to a} x = a$

3. $\lim_{x \to a} c \cdot f(x) = c \cdot \lim_{x \to a} f(x)$

4. $\lim_{x \to a} (f(x) \pm g(x)) = \lim_{x \to a} f(x) \pm \lim_{x \to a} g(x)$

5. $\lim_{x \to a} (f(x)\, g(x)) = \left(\lim_{x \to a} f(x)\right) \cdot \left(\lim_{x \to a} g(x)\right)$

6. $\lim_{x \to a} \left(\frac{f(x)}{g(x)}\right) = \frac{\lim_{x \to a} f(x)}{\lim_{x \to a} g(x)}$; $\lim_{x \to a} g(x) \neq 0$

7. $\lim_{x \to a} (f(x))^2 = \left(\lim_{x \to a} f(x)\right)^2$

Q. $\lim_{x \to 1} \frac{x^2 - 4}{x - 1}$ Does this exist?

→ NO!

Q. $\lim_{x \to 0} \frac{x}{|x|}$

→ $\lim_{x \to 0^+} = \frac{x}{x} = 1$ $\lim_{x \to 0^-} = \frac{x}{-x} = -1$. limit doesn't exist.

$\neq$

Q. $\lim_{x \to -2} (x-6)^{2/3}$

→ $= \left(\lim_{x \to -2} (x-6)\right)^{2/3}$

$= \left(\left(\lim_{x \to -2} x\right) - 6\right)^{2/3}$

$= (-2-6)^{2/3} = 4$

Q. $\lim_{x \to 2} \sqrt{x+3}$

→ $= \sqrt{\lim_{x \to 2} (x+3)}$

$= \sqrt{5}$

Q. $\lim_{y \to -3} \frac{y^2 + 4y + 3}{(y^2 - 3)}$

→ $\frac{(-3)^2 + 4(-3) + 3}{(3^2 - 3)} = 0$

Q. $f(x) = \text{int}(x)$

→ $f(x) = [x]$

$\lim_{x \to 1/2} \text{int}\, x = 0$

Q. $\lim_{x \to 1} \frac{x - 1}{x^2 - 1}$

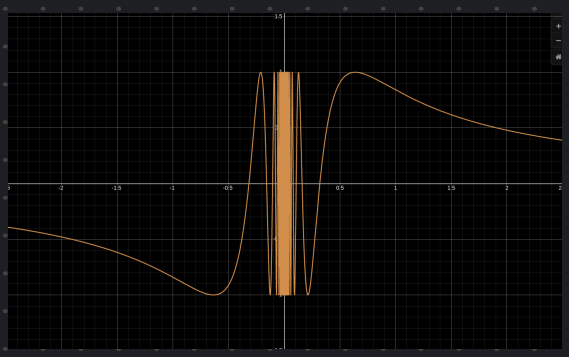
→ $= \lim_{x \to 1} \frac{(x-1)}{(x+1)(x-1)}$

$= \lim_{x \to 1} \frac{1}{(x+1)}$

$= \frac{1}{1+1} = \frac{1}{2}$

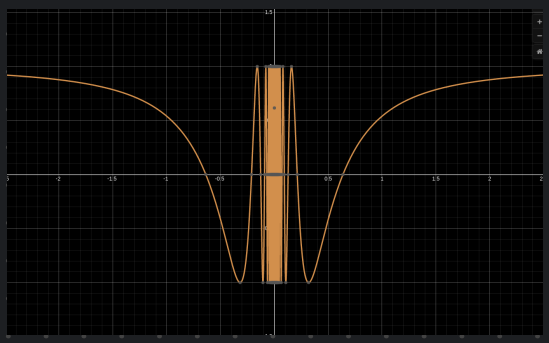
Q. $\lim_{x \to 0} \sin \frac{1}{x}$

→ doesn't exist



Q. $\lim_{x \to 0} \cos \frac{1}{x}$

→ doesn't exist



Gate questions

Q. $\lim_{x\to 0} \frac{\sin x}{x}$

→ 1

Q. $\lim_{\theta\to 0} \frac{\sin m\theta}{\theta}$

$= m \lim_{\theta\to 0} \frac{\sin m\theta}{m\theta}$

$= m \cdot 1 = 1$

Q. $\lim_{x\to 0} \frac{\sin^2 x}{x}$

$= \lim_{x\to 0} \frac{\sin x}{x} \lim_{x\to 0} \sin x$

$= 1 \cdot 0 = 0$

Q. $\lim_{\theta\to 0} \frac{\sin\theta/2}{\theta}$

$= \frac{1}{2} \lim_{\theta\to 0} \frac{\sin\theta/2}{\theta/2}$

$= 1/2$

Q. $\lim_{x\to 0} \frac{\sin x}{e^x \cdot x}$

$= \lim_{x\to 0} \frac{1}{e^x} \lim_{x\to 0} \frac{\sin x}{x}$

$= 1 \cdot 1 = 1$

Q. $\lim_{x\to 0} \frac{\cos x}{x}$ $1/0$

→ doesn't exist

Q. $\lim_{x\to 0} \frac{\tan x}{x}$ $0/0$

$= \lim_{x\to 0} \frac{\sin x}{x \cos x}$

$= \lim_{x\to 0} \frac{1}{\cos x} \lim_{x\to 0} \frac{\sin x}{x}$

$= 1 \cdot 1 = 1$

Q. $\lim_{x\to 0} \frac{\tan x}{x^2 - x}$

$= \lim_{x\to 0} \frac{\tan x}{x} \lim_{x\to 0} \frac{1}{x-1}$

$= 1 \cdot -1 = -1$

Q. $\lim_{x\to 0} \frac{\sin 4x}{\sin 2x}$

$= \lim_{x\to 0} \frac{\sin 4x}{x} \frac{x}{\sin 2x}$

$= 4 \lim_{x\to 0} \frac{\sin 4x}{4x} \frac{1}{2} \lim_{x\to 0} \frac{2x}{\sin 2x}$

$= 4 \cdot \frac{1}{2} = 2$

$\lim_{x\to 0} \frac{x^3 + x^2}{2x^3 - 7x^2}$ $0/0$

$= \lim_{x\to 0} \frac{x+1}{2x - 7}$

$= \frac{-1}{7}$

Q. $\lim_{x\to\infty} \frac{\sin x}{x}$

(sin∞ would be in $[-1, 1]$

$= \frac{\text{finite}}{\infty} \to 0$

Q. $\lim_{x\to\infty} \left(\sqrt{x^2+x} - \sqrt{x^2+1}\right)$

$= \lim_{x\to\infty} \left(\sqrt{x^2+x} - \sqrt{x^2+1}\right) \times \frac{\sqrt{x^2+x} + \sqrt{x^2+1}}{\sqrt{x^2+x} + \sqrt{x^2+1}}$

$= \lim_{x\to\infty} \frac{x^2+x-x^2-1}{\sqrt{x^2+x} + \sqrt{x^2+1}}$

$= \lim_{x\to\infty} \frac{x-1}{\sqrt{x^2+x} + \sqrt{x^2+1}}$

$= \lim_{x\to\infty} \frac{1 - \frac{1}{x}}{\sqrt{1+\frac{1}{x}} + \sqrt{1+\frac{1}{x^2}}}$

$= \frac{1-0}{\sqrt{1+0} + \sqrt{1+0}} = \frac{1}{2}$

Q. $\lim_{x\to\infty} \frac{x^2 - 5x + 4}{4x^2 + 2x}$

$= \lim_{x\to\infty} \frac{1 - \frac{5}{x} - \frac{4}{x^2}}{4 + \frac{2}{x}}$

$= 1/4$

Q. $\lim_{x\to\infty} \frac{\sqrt{9x^2 + 2020}}{x + 7}$

$= \lim_{x\to\infty} \frac{\sqrt{9 + \frac{2020}{x^2}}}{1 + 7/x}$

$= 3$

Q. $\lim_{x\to\infty} \frac{x - \sin x}{x + \cos x}$

$= \lim_{x\to\infty} \frac{1 - \frac{\sin x}{x}}{1 + \frac{\cos x}{x}}$

$= \frac{1-0}{1+0} = 1$

$\lim_{x\to 5} g(x) =$ ____

$\lim_{x\to 5} f(x) = 2$

$\lim_{x\to 5} h(x) = 2$

→ $f(x) \le g(x) \le h(x)$ over the interval I.

a) Doesn't exist

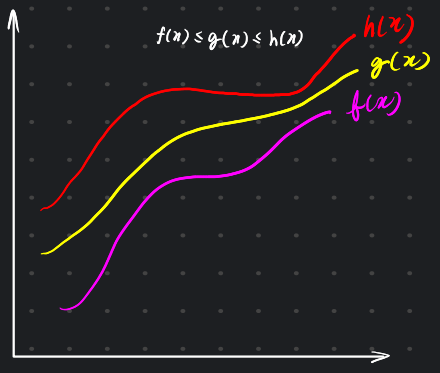
b) 1/2

c) 2

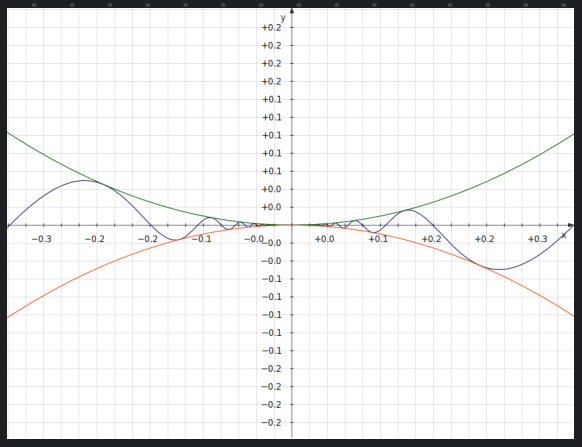
d) 0

→ 2 (squeeze theorem)

Squeeze theorem/ Sandwich theorem:



If $\lim_{x \to a} f(x) = L$ and $\lim_{x \to a} h(x) = L$ then $\lim_{x \to a} g(x) = L$



Theorem — Let I be an interval containing the point a. Let g, f, and h be functions defined on I, except possibly at a itself. Suppose that for every x in I not equal to a, we have

$$g(x) \leq f(x) \leq h(x)$$

and also suppose that

$$\lim_{x \to a} g(x) = \lim_{x \to a} h(x) = L.$$

Then $\lim_{x \to a} f(x) = L$.

Q. $\lim_{x \to 0} x^2 \sin \frac{1}{x}$

$\longrightarrow$ $-1 \leq \sin \frac{1}{x} \leq 1$

$-x^2 \leq x^2 \sin x \leq x^2$

$\lim_{x \to 0}(-x^2) = 0$ $\quad$ $\lim_{x \to 0}(x^2) = 0$

$\therefore$ By squeeze theorem,

$\lim_{x \to 0} x^2 \sin x = 0$

Q. $\lim_{x \to 0} x \sin \frac{1}{x}$

$\longrightarrow$ $-1 \leq \sin \frac{1}{x} \leq 1$

$-x \leq x \sin x \leq x$

$\downarrow \lim_{x \to 0}(-x)$ $\quad$ $\downarrow \lim_{x \to 0} x$

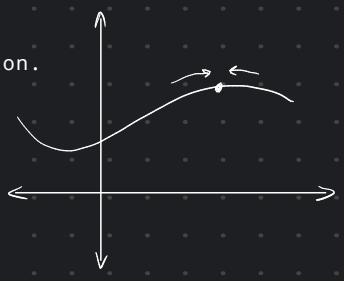
0 $\quad$ 0

$\therefore \lim_{x \to 0} x \sin x = 0$

Continuity

In mathematics, a continuous function is a function such that
a small variation of the argument induces a small variation of the value of the function.
This implies there are no abrupt changes in value, known as discontinuities.

A real function is continuous if the graph is a single unbroken curve
whose domain is the entire real line.



Continuity of real functions is usually defined in terms of limits.
A function f with variable x is continuous at the real number c,
if the limit of f(x), as x tends to c, is equal to f(c).

At any a,

$$f(a) = \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x) = \lim_{x \to a} f(x)$$

eg.

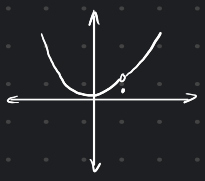
$$f(x) = \begin{cases} 5, & x > 0 \\ -5, & x < 0 \\ 0, & x = 0 \end{cases}$$



$\rightarrow \lim_{x \to 0^-} f(x) = -5 \quad \lim_{x \to 0^+} f(x) = 5$

Discontinuous

eg. $f(x) = \begin{cases} x^2 ; & x \neq 2 \\ 1 ; & x = 2 \end{cases}$



$\rightarrow \lim_{x \to 2^-} f(x) = 4 \quad \lim_{x \to 2^+} f(x) = 4$

$\therefore \lim_{x \to 2} f(x) = 4$

But $f(2) = 1 \neq \lim_{x \to 2} f(x)$

$\therefore f(x)$ is discontinuous at $x = 2$

Q. $f(x) = \begin{cases} \frac{x^2 - 9}{x - 3} ; & \text{if } x \neq 3 \\ a ; & \text{if } x = 3 \end{cases}$ If $f(x)$ is continuous, find a.

$\rightarrow f(a) = \lim_{x \to a^-} f(x) = \lim_{x \to a^+} f(x)$

$\lim_{x \to 3} \frac{x^2 - 9}{x - 3} = \lim_{x \to 3} \frac{(x+3)(x-3)}{(x-3)}$

$= \lim_{x \to 3} (x+3) = 6$

$f(3) = 6 \quad a = 6$

Q. $f(x) = \begin{cases} e^x ; & x < 0 \\ 9x^2 + x + 1 ; & x \geq 0 \end{cases}$

$\rightarrow f(0) = 1$

$\lim_{x \to 0^-} f(0) = \lim_{x \to 0^-} e^x = 1$

$\lim_{x \to 0^+} f(0) = \lim_{x \to 0^+} (9x^2 + x + 1) = 1$

$\therefore$ continuous

Q. $f(x) = |x|$. continuous at $x = 0$?

$\rightarrow \lim_{x \to 0^-} |x| = \lim_{x \to 0^-} (-x) = 0$ $\therefore$ continuous

$\lim_{x \to 0^+} |x| = \lim_{x \to 0^+} (x) = 0$

$f(0) = 0$

Properties of continuity:

- If f and g are continuous functions at x=c, then the following are continuous as well:
  - f+g at x=c
  - f-g at x=c
  - f*g at x=c
  - kf at x=c (k is a constant)
  - f/g at x=c (g(c) != 0)

- If f is continuous at c, g is continuous at f(c), then the composite gof is continuous at c.

e.g. $\left| \frac{x \sin x}{x^2+2} \right|$

$x \to$ cont.
$\sin x \to$ cont.
$\therefore x \sin x \to$ cont.

$x^2 \to$ cont.
$\frac{x \sin x}{x^2+2} \to$ cont.

$f(x) = \frac{x \sin x}{x^2+2}$ is continuous.

$g(x) = |x|$ is continuous.

$\therefore g(f(x))$ is continuous.

Derivative at a point

$$\boxed{f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}}$$

let $h = x - a \Rightarrow x = a + h$

when $x \to a$, $x - a \to 0$, $h \to 0$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

$$\boxed{f'(x) = \lim_{h \to 0} \frac{(x+h) - f(x)}{h}}$$

e.g. Derivative of x^3 ?

→
$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$= \lim_{h \to 0} \frac{(x+h)^3 - x^3}{h}$$

$$= \lim_{h \to 0} \frac{3h^2x + 3hx^2}{h}$$

$$= \lim_{h \to 0} 3hx + 3x^2$$

$$= 3x^2$$

Differentiability in an interval

f(x) is said to be differentiable over the interval (a, b) if it is differentiable at every point in (a, b).

f(x) is said to be differentiable over the interval [a, b] if it is:
- differentiable at every point in (a, b),
- right derivative exists at x=a, and left derivative exists at x=b. (left and right slopes)

Q. $f(x) = |x|$

a) Is this differentiable over (2, 10)? ✓ It is not differentiable at $x=0$.

b) over (-5, -2)? ✓

c) over (-7, 5)? ✗

Q. Given $f(x) = \begin{cases} x^2, & x \le 0 \\ 2x, & x > 0 \end{cases}$ Does its derivative exist at $x=0$?



→ $f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$

$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$

Left derivative:

$\lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$

$= \lim_{h \to 0} \frac{(a+h)^2 - a^2}{h}$

$= \lim_{h \to 0} \frac{h^2 + 2ah}{h}$

$= \lim_{h \to 0} h + 2a$

$= 0$

Right derivative:

$\lim_{h \to 0} \frac{2(a+h) - 2a}{h}$

$= \lim_{h \to 0} \frac{2h}{h}$

$= 2$

∴ The derivative doesn't exist at $x=0$.

Q. $f(x) = \frac{1}{x}$, find $f'(a)$ at $a=2$.

→ $f'(a) = \lim_{x \to a} \frac{f(x) - f(a)}{x - a}$

$= \lim_{x \to a} \frac{\frac{1}{x} - \frac{1}{a}}{x - a}$

$= \lim_{x \to a} \frac{-(x-a)}{xa} \times \frac{1}{(x-a)}$

$= -\frac{1}{a^2}$

$f'(2) = -\frac{1}{4}$.

Standard derivatives

$\frac{d(k)}{dx} = 0$

$\frac{d(x)}{dx} = 1$

$D(kx) = k$

$D(x^n) = n x^{n-1}$

$D(e^x) = e^x$

$D(\ln x) = \frac{1}{x}$

$D(a^x) = a^x \log a$

$D(x^x) = x^x (1 + \ln x)$

$D(\log_a x) = \frac{1}{x} \frac{1}{\ln a}$

$D(\sin x) = \cos x$

$D(\cos x) = -\sin x$

$D(\tan x) = \sec^2 x$

$D(\cot x) = -\operatorname{cosec}^2 x$

$D(\sec x) = \sec x \tan x$

$D(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$

$D(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$

$D(\cos^{-1} x) = \frac{-1}{\sqrt{1-x^2}}$

$D(\tan^{-1} x) = \frac{1}{1+x^2}$

$D(\cot^{-1} x) = \frac{-1}{1+x^2}$

$D(\sec^{-1} x) = \frac{1}{|x| \cdot \sqrt{x^2-1}}$

$D(\operatorname{cosec}^{-1} x) = \frac{-1}{x\sqrt{x^2-1}}$

- $\frac{d}{dx}.x^n = n.\ x^{n-1}$
- $\frac{d}{dx}.\ k = 0$, where k is a constant
- $\frac{d}{dx}.e^x = e^x$
- $\frac{d}{dx}.a^x = a^x.\ \log_e .a$, where $a > 0, a \neq 1$
- $\frac{d}{dx}.\log x = 1/x,\ x > 0$
- $\frac{d}{dx}.\ \log_a e = 1/x \log_a e$
- $\frac{d}{dx}.\sqrt{x} = 1/(2\sqrt{x})$

- $\frac{d}{dx}.\sin x = \cos x$
- $\frac{d}{dx}.\cos x = -\sin x$
- $\frac{d}{dx}.\tan x = \sec^2 x$, $x \neq (2n+1)\ \pi/2$, $n \in I$
- $\frac{d}{dx}.\ \cot x = -\ \mathrm{cosec}^2 x,\ x \neq n\pi,\ n \in I$
- $\frac{d}{dx}.\ \sec x = \sec x \tan x,\ x \neq (2n+1)\ \pi/2$, $n \in I$
- $\frac{d}{dx}.\mathrm{cosec}\ x = -\ \mathrm{cosec}\ x \cot x,\ x \neq n\pi,\ n \in I$

- $\frac{d}{dx}.\ \sin^{-1} x = \frac{1}{\sqrt{(1-x^2}}$, $-1 < x < 1$
- $\frac{d}{dx}.\cos^{-1} x = -\frac{1}{\sqrt{(1-x^2}}$, $-1 < x < 1$
- $\frac{d}{dx}.\ \tan^{-1} x = \frac{1}{(1+x^2)}$
- $\frac{d}{dx}.\ \cot^{-1} x = -\frac{1}{(1+x^2)}$
- $\frac{d}{dx}.\ \mathrm{cosec}^{-1} x = -\frac{1}{|x|\sqrt{x^2-1}}$, $|x| > 1$

- $\frac{d}{dx}.\ \sinh x = \cosh x$
- $\frac{d}{dx}.\ \cosh x = \sin hx$
- $\frac{d}{dx}.\ \tan hx = \sec h^2 x$
- $\frac{d}{dx}.\ \cot hx = -\mathrm{cosec}\ h^2 x$
- $\frac{d}{dx}.\ \sec hx = -\mathrm{sech}\ hx \tan hx$
- $\frac{d}{dx}.\ \mathrm{cosec}\ hx = -\mathrm{cosec}\ hx \cot hx$

- $\frac{d}{dx}.\ \sinh^{-1} x = \frac{1}{\sqrt{(x^2+1)}}$
- $\frac{d}{dx}.\cosh^{-1} x = -\frac{1}{\sqrt{(x^2-1)}}$
- $\frac{d}{dx}.\ \tanh^{-1} x = \frac{1}{(1-x^2)}$
- $\frac{d}{dx}.\ \coth^{-1} x = -\frac{1}{x(1-x^2)}$
- $\frac{d}{dx}.\ \mathrm{cosech}^{-1} x = -\frac{1}{x\sqrt{1+x^2}}$

Derivatives of Common Functions

In this table, a is a constant, while u, v, w are functions. The derivatives are expressed as derivatives with respect to an arbitrary variable x.

$$\frac{d}{dx}(a) = 0$$

$$\frac{d}{dx}(x) = 1$$

$$\frac{d}{dx}(au) = a\frac{du}{dx}$$

$$\frac{d}{dx}(u + v - w) = \frac{du}{dx} + \frac{dv}{dx} - \frac{dw}{dx}$$

$$\frac{d}{dx}(uv) = u\frac{dv}{dx} + v\frac{du}{dx}$$

$$\frac{d}{dx}\left[\frac{u}{v}\right] = \frac{1}{v}\frac{du}{dx} - \frac{u}{v^2}\frac{dv}{dx}$$

$$\frac{d}{dx}(u^n) = nu^{n-1}\frac{du}{dx}$$

$$\frac{d}{dx}(\sqrt{u}) = \frac{1}{2\sqrt{u}}\frac{du}{dx}$$

$$\frac{d}{dx}\left[\frac{1}{u}\right] = -\frac{1}{u^2}\frac{du}{dx}$$

$$\frac{d}{dx}\left[\frac{1}{u^n}\right] = -\frac{n}{u^{n+1}}\frac{du}{dx}$$

$$\frac{d}{dx}[f(u)] = \frac{d}{du}[f(u)]\frac{du}{dx}$$

$$\frac{d}{dx}[\ln u] = \frac{d}{dx}[\log_e u] = \frac{1}{u}\frac{du}{dx}$$

$$\frac{d}{dx}[\log_a u] = \log_a e\frac{1}{u}\frac{du}{dx}$$

$$\frac{d}{dx}e^u = e^u\frac{du}{dx}$$

$$\frac{d}{dx}a^u = a^u \ln a\frac{du}{dx}$$

$$\frac{d}{dx}\left(u^v\right) = vu^{v-1}\frac{du}{dx} + \ln u \;\; u^v\frac{dv}{dx}$$

$$\frac{d}{dx}\sin u = \cos u\frac{du}{dx}$$

$$\frac{d}{dx}\cos u = -\sin u\frac{du}{dx}$$

$$\frac{d}{dx}\tan u = \sec^2 u\frac{du}{dx}$$

$$\frac{d}{dx}\cot u = -\csc^2 u\frac{du}{dx}$$

$$\frac{d}{dx}\sec u = \sec u\tan u\frac{du}{dx}$$

$$\frac{d}{dx}\csc u = -\csc u\cot u\frac{du}{dx}$$

Rules

Product rule:

$$(uv)' = uv' + u'v$$

Quotient rule:

$$\left(\frac{u}{v}\right)' = \frac{vu' - uv'}{v^2}$$

NOTE: every differentiable func^n is continuous; may not be true the other way around.

Given $f(x)$ is diff^{ble}, prove $\lim_{x \to a} f(x) = f(a)$.

→ LHS $= \lim_{x \to a} f(x)$ RHS $= f(a)$

differentiable → continuous

∿ continuous ⟶ ∿ differentiable

Q. $f(x) = x \ln x - x$. Find $f'(x)$ for $x > 0$

→ $f'(x) = x\left(\frac{1}{x}\right) + (1)\ln x - 1$

$= \ln x$

Q. Find k if $f(x)$ is differentiable at $x = 1$. $f(x) = \begin{cases} x^3, & x \le 1 \\ 3x + k, & x > 1 \end{cases}$

→ Diff → continuous

$\lim_{x \to 1} f(x) = f(1) = 1$

$\lim_{x \to 1^-} f(x) = 1$ $\lim_{x \to 1^+} f(x) = 3 + k$ $3 + k = 1$

$k = -2$

03. Given the following statements about a function f: R→R, select the right option:
P : If f(x) is continuous at $x = x_0$, then it is also differentiable at $x = x_0$
Q : If f(x) is continuous at $x = x_0$, then it may not be differentiable at $x = x_0$
R : If f(x) is continuous at $x = x_0$, then it is also different at $x = x_0$
(A) P is true, Q is false, R is false (B) P is false, Q is true, R is true
(C) P is false, Q is true, R is false (D) P is true, Q is false, R is true

✓ B

Which of the following functions is not differentiable in the domain $[-1, 1]$?

(a) $f(x) = x^2$

(b) $f(x) = x - 1$

(c) $f(x) = 2$

(d) $f(x) = Maximum(x, -x)$

D ✓

Indeterminate forms

I. $\frac{0}{0}$ ← mother of all

II. $0 \cdot \infty$ $\quad \frac{0}{0} = 0 \cdot \frac{1}{0} = 0 \cdot \infty$

III. $\frac{\infty}{\infty}$ $\quad ab = \frac{b}{\frac{1}{a}} = \frac{a}{\frac{1}{b}}$

IV. $\infty - \infty$ $\quad \frac{0}{0} = \frac{1-1}{0} = \infty - \infty$

V. ∞^0 $\quad \frac{\infty^1}{\infty^1} = \infty^{1-1} = \infty^0$

VI. 0^0

VII. 1^∞ $\quad 1^\infty = \left(\frac{2}{2}\right)^\infty = \frac{2^\infty}{2^\infty} = \frac{\infty}{\infty}$

so 1^∞ is indeterminate as well. example: $\lim_{n \to \infty} \underbrace{\left(1 + \frac{1}{n}\right)^n}_{\left(1+\frac{1}{\infty}\right)^\infty \left(1+0\right)^\infty 1^\infty} = e$

Q. 0^∞ ?

→ $= 0$ not an indeterminate form.

Q. $0^{-\infty}$

→ $= \frac{1}{0^\infty}$

$= \frac{1}{0}$

$= \infty$ not indeterminate

Q. $\lim_{n \to \infty} \frac{x^2}{x}$ $\quad \lim_{n \to \infty} \frac{x}{x^2}$ $\quad \lim_{n \to \infty} \frac{x}{x}$

$= \lim_{n \to \infty} x$ $\quad = \lim_{n \to \infty} \frac{1}{x}$ $\quad = 1$

$= \infty$ $\quad = 0$

L'Hôpital's rule L'Hôpital's Rule

De L'Hôpital's rule states that for functions f and g which are differentiable on an open interval I except possibly at a point c contained in I, if $\lim_{x\to c} f(x) = \lim_{x\to c} g(x) = 0$ or $\lim_{x\to c} g(x) = \pm\infty$, and $g'(x) \neq 0$ for all x in I with $x \neq c$, and $\lim_{x\to c} \frac{f'(x)}{g'(x)}$ exists, then

$$\lim_{x\to c} \frac{f(x)}{g(x)} = \lim_{x\to c} \frac{f'(x)}{g'(x)}.$$

e.g. $\lim_{x\to 0} \frac{\sin x}{x} = \lim_{x\to 0} \frac{\cos x}{1} = 1$

Proof:

$\lim_{x\to a} \frac{f(x)}{g(x)}$ say $f(a)=0$, $g(a)=0$ $\left[\frac{0}{0}\text{ form}\right]$

$= \lim_{x\to a} \frac{f(x)-0}{g(x)-0}$

$= \lim_{x\to a} \frac{f(x)-f(a)}{g(x)-g(a)}$

$= \lim_{x\to a} \frac{f(x)-f(a)}{x-a} \quad \frac{x-a}{g(x)-g(a)}$

$= \lim_{x\to a} \frac{f'(x)}{g'(x)}$

3.